

What Makes a Video Game a “Top-Seller”

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The global video game business is a multi-billion dollar industry and has been for many years. As technology improves, so does gameplay, which in turn, encourages further profit potential. In 2024 the revenue from the global gaming market was estimated at almost 455 billion U.S. dollars. (cite statista) Despite the large revenue and profit potential of video games in general, a number of game releases fail to achieve commercial success. Game developers and publishers would benefit from the ability to predict which games will perform well, before launch, and greatly improve decisions regarding investments, marketing, reducing risk, and maximizing profitability. The goal of this analysis is to investigate whether we could build a predictive model using global sales data to identify top-selling games.

Data

The dataset, hosted on Kaggle, aggregates video game sales and ratings based on Metacritic scores and provides insight into global video game sales and ratings. Initially, the dataset contained over 16,000 entries, after cleaning and removing rows with excessive missing values, we retained 6,825 observations. The decision to not impute missing values was made primarily because this dataset was comprised of two datasets that were merged. Imputation may have introduced more bias and the goal was to focus on applying machine learning models. The focus was primarily on features available prior to release; platform, year of release, genre, publisher, developer, critic/user scores, and rating. The outcome variable Top_Seller_75 was engineered to classify games that ranked in the top 25% of global sales.

Methods

We utilized two supervised machine learning models, logistic regression and random forest. The dataset was split into training and testing sets (80/20). To ensure the reliability and generalizability of our results, we implemented a 5-fold cross-validation strategy during model training. The analysis

pipeline began with comprehensive feature encoding and selection to prepare the dataset and highlight the most informative predictors. Following this, we trained and tested multiple classification models, evaluating their performance using a range of key metrics, including accuracy, precision, recall, and F1 score. In addition to these standard metrics, we conducted a ROC-AUC curve analysis to assess the models' ability to distinguish between classes across various threshold settings. To further examine model performance, we reviewed the confusion matrix, which provided insight into the distribution of true positives, false positives, true negatives, and false negatives. Finally, to optimize model performance, we conducted hyperparameter tuning using GridSearchCV, allowing us to systematically explore and identify the best combination of parameters for each model. All data cleaning, analysis, and modeling were performed using Python, utilizing libraries such as pandas for data manipulation, seaborn and matplotlib for visualization, and scikit-learn for machine learning algorithms and evaluation metrics.

Results Discussion

Exploratory data analysis showed extensive variation in global video game sales. On average, games in the processed dataset sold about 0.78 million units, but the distribution was heavily right skewed. Most games had fairly low sales, while a small number sold tens of millions, with a few outliers even surpassing 80 million units. The outliers weren't removed when judged to be valid blockbuster hits that dominated the global market. Figure 1 illustrates the distribution of global sales, limited to games that sold fewer than 5 million units, to better highlight the concentration of lower-selling titles without the distortion caused by extreme outliers.

When we looked at which genres were most likely to produce top sellers, Misc, Shooter, and Sports stood out, making up 32%, 31%, and 27% of the top-performing titles respectively. That said, Action games were by far the most common genre overall, with 1,630 titles in the dataset, followed by

Sports and Shooter games. This pattern is captured in Figure 2, which shows the proportion of top sellers by genre, offering a clearer view of genre-based performance trends in the market.

I started with a basic logistic regression model and used cross-validation to evaluate how well it could classify top sellers. The model performed reasonably well, with precision and recall in the range of 0.65 to 0.70, suggesting a reasonable balance between identifying popular games and minimizing false positives. Figure 3 presents the ROC curve for the logistic regression model, showing an AUC of 0.80, which indicates solid discriminatory power, though not optimal.

When I switched to a random forest classifier and tuned the hyperparameters using GridSearchCV, performance improved substantially. These tree-based models were better at capturing more complex patterns in the data and interactions between variables, which helped push the F1 score closer to 0.75. Looking at the confusion matrix for the tuned random forest model (Figure 4), we can see that most of the model's misclassifications occurred in borderline cases—games that just barely made or missed the threshold for being considered a top seller.

Genre and platform were among the most commonly assumed predictors of sales success, but the random forest model revealed that user and critic engagement metrics were actually more influential. As shown in the feature importance plot (Figure 5), User_Count, Critic_Score, and Critic_Count ranked highest—outweighing genre, publisher, and even platform in predictive power. This finding suggests that how a game is received, even before or shortly after release, is more telling of its commercial success than its category or release platform.

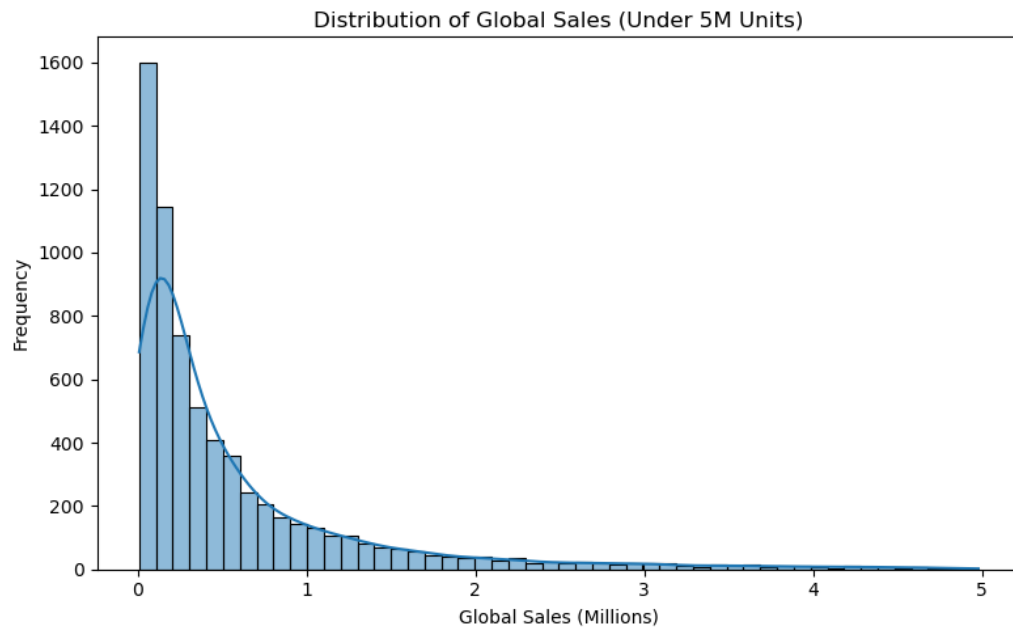
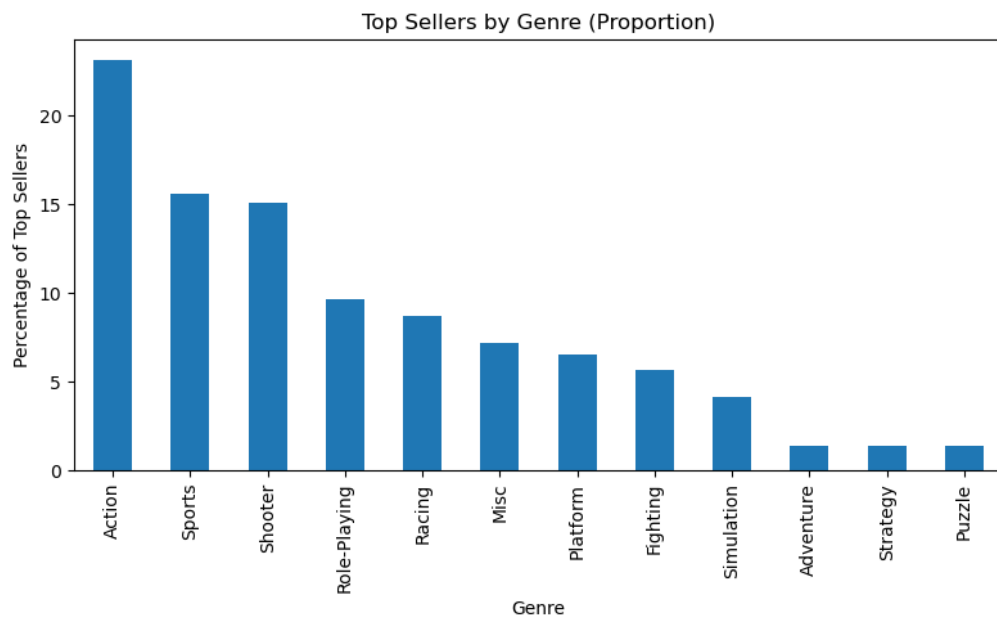
Figure 1 Distribution of Global Sales (Games Under 5 Million Units)*Figure 2 Top Sellers by Genre (Proportion)*

Figure 3 ROC Curve for Logistic Regression

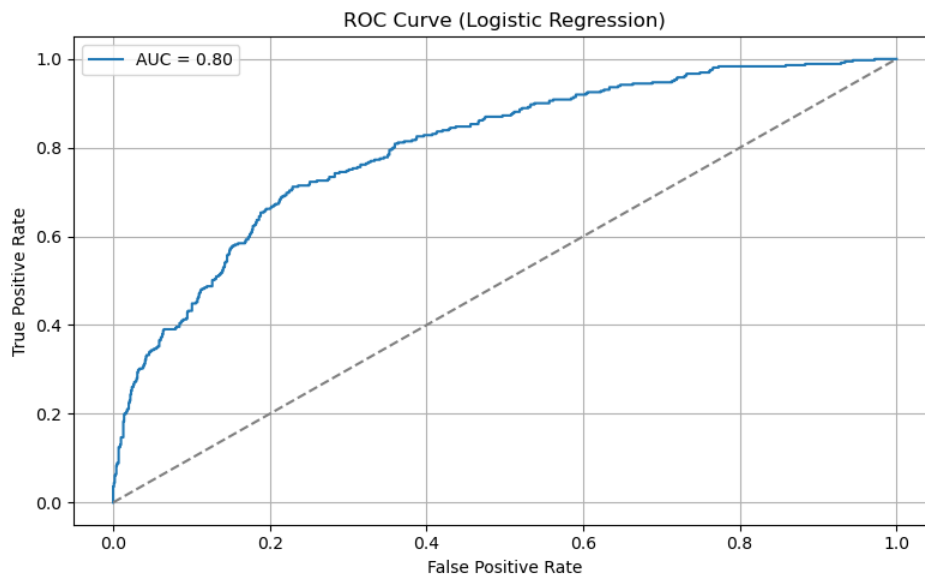


Figure 4 Confusion Matrix (Tuned Random Forest Classifier)

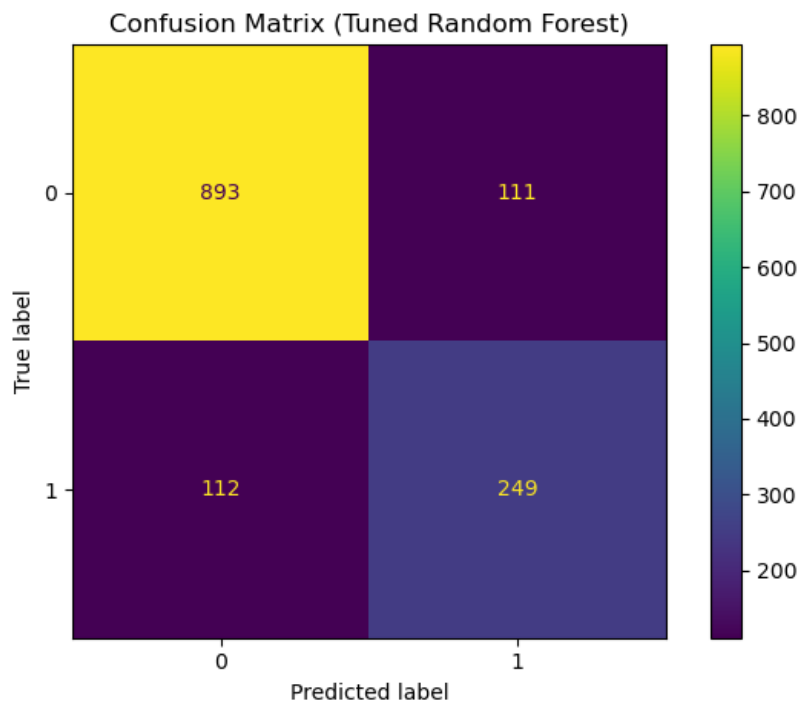
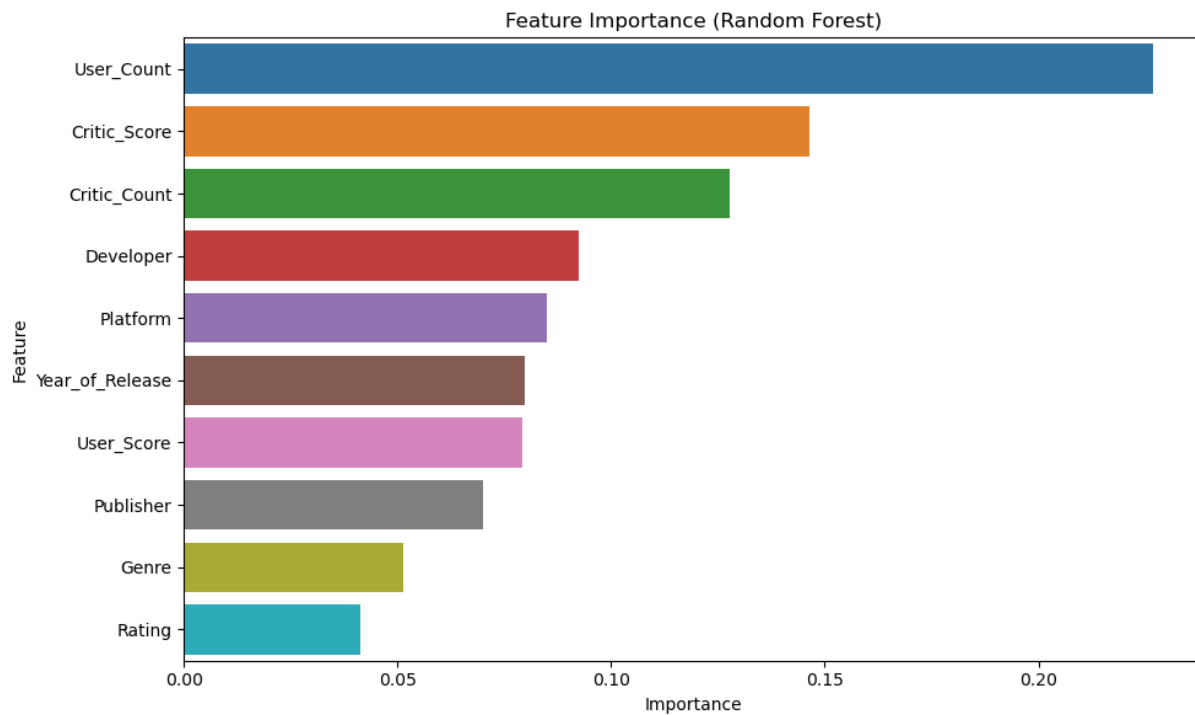


Figure 5 Feature Importance from Random Forest Model

Strengths, Limitations, and Future Work

A key strength of this project was the use of real-world publicly available data from two merged surveys, which offered a diverse range of games and features for analysis. The application of multiple machine learning models and the use of cross-validation and grid search helped ensure robust performance comparisons.

However, the project also faced limitations. The exclusion of rows with missing values significantly reduced the dataset size, which may have impacted model generalizability. Additionally, the dataset lacked important predictors like marketing budgets, release timing, and user reviews—factors that can strongly influence game sales. Another key limitation was the historical nature of this dataset, the video game industry is constantly evolving and with trend shifting in the industry, model performance may drop.

For future work, incorporating additional data sources, such as social media sentiment or review metadata, could enhance prediction accuracy. Experimenting with advanced models like gradient boosting or neural networks, and performing time-based splits to mimic real forecasting conditions, would also be valuable next steps.

Conclusion

This project helped uncover what makes a video game a top seller—and the answer isn't always simple. Sales were heavily skewed, with most games selling modestly while a few blockbusters dominated the market. Genre made a big difference. Miscellaneous, Shooter, and Sports titles were especially common among top performers, though Action games showed up the most overall.

I started with logistic regression, which gave decent results and helped set a baseline, but it became clear that it wasn't quite capturing the complexity of the data. Switching to a random forest model and tuned it using GridSearchCV, performance improved significantly, with a stronger F1 score and more accurate predictions. The tuned model still struggled in borderline cases—games that were close to the threshold for “top seller”—but overall, it performed reliably.

What really stood out from the feature importance plot was that user and critic engagement were strong indicators of success. Things like User_Count and Critic_Score were more important than even genre or publisher, which shows how much buzz and reception matter in the gaming world.

This whole process reinforced how valuable it is to explore the data deeply, test different models, and experiment with tuning. It wasn't just about getting a high accuracy score—it was about understanding what drives game success and how those insights can help future decisions in game development or marketing.

References

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